

Scene / Chapter	Narration (Voice or On-Screen Text)	Visual Cue (What Students See)	Action (Easy Animation Steps)
Opening Slide (Title Scene)	"In a digital world where data holds everything together... one young coder, Algo, discovers a secret path — the Land of Data." "Each door he opens reveals how data lives, moves, and grows."	Black background → digital grid forms, binary flows. Title text fades in: "Algo's Journey Through the Land of Data (DSA)"	Title fade-in/out. Stickman (Algo) appears. 12 doors appear around him. Door 1 glows. Algo looks curiously.
1. Introduction to Data Structures & Searching	"Every structure begins with data. But to find what we need, we must search ." "Linear Search checks one by one. Binary Search splits the data to find faster."	Row of boxes (array). Target marked. Split-screen for binary search.	Algo walks checking each box for linear search. Binary search shows middle halving. Target turns green. Door 2 glows.
2. Arrays & Sparse Matrices	"Arrays are like neat lockers — fixed and continuous." "But when most lockers are empty, we compress and store only useful ones — Sparse Matrices."	A matrix with mostly empty cells → highlight few filled ones. Smaller 3-box sparse structure appears.	Algo shakes head at wasted empty space. Only non-zero values remain in compact box format. Door 3 glows.
3. Stack & Its Applications	"Some data grows like a tower — the last added is the first removed." "This is the Stack ."	Tower of boxes labeled data elements.	Boxes slide up (Push) and pop off (Pop). A small expression transforms from infix → postfix. Door 4 glows.
4. Queue & Its Variants	"Queues follow the rule of waiting in line — First In, First Out." "Circular queues loop back. Deques allow both ends."	Straight line of boxes → arrows showing movement. Circular ring queue. Deque with two ends.	Boxes enter from rear, exit from front. Circular queue rotates. Deque shows insertion/removal from both sides. Door 5 glows.
5. Linked Lists & Memory Management	"Linked Lists are flexible — they grow and shrink inside scattered memory."	Boxes labeled Data + Next pointer arrows. Doubly linked arrows. Circular linking.	Algo draws arrows connecting nodes, inserts one node between two. Memory block appears showing malloc/free. Door 6 glows.

6. Trees & Binary Trees	“Trees branch out like families. Each node splits into children.” “Binary Trees have two special links — Left & Right.”	Root node appears → two nodes below → expand into tree shape.	Algo walks three paths one by one: Preorder, Inorder, Postorder while nodes highlight. Door 7 glows.
7. Advanced Trees (BST, AVL, B+ Trees Basic)	“Binary Search Trees keep data sorted.” “AVL Trees balance themselves.” “B+ Trees store many keys for faster storage access.”	BST growing uneven → imbalance. AVL rotation fixes it. Wide B+ tree block appears like a filing cabinet.	Algo pushes heavy side of tree — rotation animation. Keys in block align neatly. Door 8 glows.
8. Graphs & Their Representations	“Welcome to the network — the world of Graphs.” “Everything connects: cities, friends, devices.”	Node-link network appears. Two views side-by-side: Matrix table vs List diagram.	Touching a node highlights neighbors. Matrix cells light up. List pointers glow. Door 9 glows.
9. Graph Traversal (DFS & BFS)	“To explore a graph, we can dive deep or spread wide.” “DFS goes deep. BFS explores level by level.”	Same graph continues. DFS path. BFS expanding layers.	Algo walks along deep single path → nodes glow in order (DFS). BFS lights nodes layer-by-layer like ripples. Door 10 glows.
10. Sorting Algorithms (Part 1)	“Sorting brings order to chaos.” “Simple sorts: Bubble, Insertion, Selection.”	Numbers float randomly → align after sorts.	Bubble: swaps upward. Insertion: one element moves each step. Selection: smallest found and swapped. Door 11 glows.
11. Sorting Algorithms (Part 2)	“Faster sorting — divide, conquer, and organize.” “Merge, Quick, Heap, and Radix Sort help handle big data.”	Split & merge blocks, pivot partitions, heap pyramid, number buckets.	Merge Sort halves/rejoins. Quick Sort shows pivot split. Heap reshuffles pyramid. Radix drops numbers into digit buckets. Door 12 glows brightly.
12. Hashing & Collision Handling	“When we want instant access — we don’t search... we calculate.” “Hashing jumps directly to the correct place.” “Even if two collide, chaining and probing resolve it fast.”	A Hash Machine in center, array of slots below.	Data enters machine → teleports into correct slot. Collision: new node linked in chain. Probing: key slides to next free slot. All 12 doors merge into a large shining door.

Closing Scene – The End	“Algo has learned every secret — from arrays to hashing.” “He restored order to data... and found the true structure of logic.” “Every bit of data hides a story — and this was his.”	All data worlds merge into one bright 3D structure: Data Landscape. Text: “The End — or The Beginning?”	Algo waves goodbye, walks into glowing door. Screen fades to white, then black.
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